

Visualizing Multilevel Models

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2026-09-21

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1 Introduction

An evolving set of notes on visualizing results from multilevel models.

The examples below use the `simulated_multilevel_data.dta` file from my text book on *Multilevel Thinking*. Here is a [link to download the data](#).

This document relies on the extraordinary `Statamarkdown` library (Hemken 2023).

2 Organizing Questions

Try to think about some of the advantages and disadvantages of different approaches to visualizing multilevel models. In multilevel models, we don't want to just *control for* variation, but to start to *explore* the variation. Put concretely:

- Some approaches use *dots*. Some approaches use *lines*. Some approaches use *dots and lines*.
- Some approaches use the *raw unadjusted* data. Other approaches use *adjusted or model predicted* data.
- Some approaches attempt to show the *Level 2 specific regression lines*; some approaches only show an *average regression line*.
- What approaches might work well with *large numbers* of Level 2 units? What approaches might work well with *smaller numbers* of Level 2 units?

What approach(es) do you prefer?

3 Setup

I am not terrifically fond of the default `s2color` graph scheme in earlier versions of Stata. Here I make use of the `michigan` graph scheme available at: <https://agrogan1.github.io/Stata/michigan-graph-scheme/>.

```
set scheme michigan
```

Stata's `stcolor` scheme—available in newer versions of Stata—would also be an option as would be Asjad Naqvi's incredible `schemepack`: <https://github.com/asjadnaqvi/stata-schemepack>.

Throughout the tutorial, I make frequent use of the `mcolor(%30)` option to add some visual interest to scatterplots by adding transparency to the markers.

4 Get Data

```
use "simulated_multilevel_data.dta", clear
```

5 Scatterplots (twoway scatter y x)

```
twoway scatter outcome warmth, mcolor(%30)  
graph export myscatter.png, width(1500) replace
```

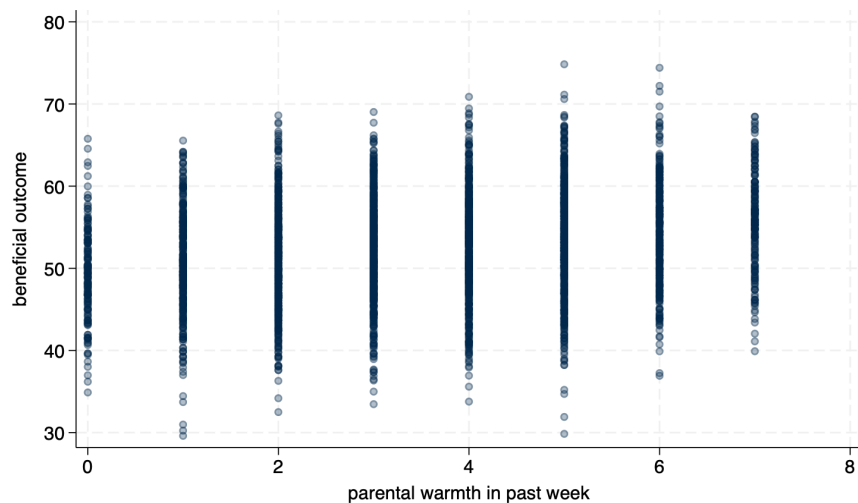


Figure 1: Scatterplot

6 Simple Linear Fit (twoway lfit y x)

```
twoway lfit outcome warmth  
  
graph export mylinear.png, width(1500) replace
```

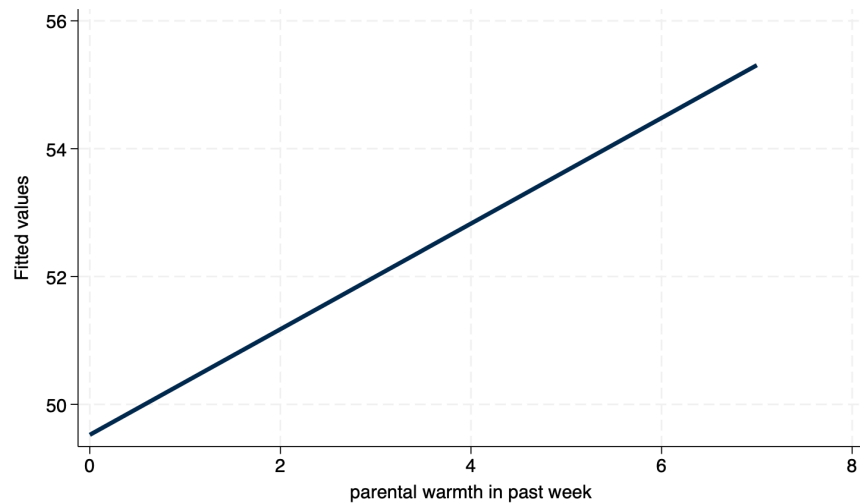


Figure 2: Linear Fit

7 Linear Fit With Confidence Interval (twoway lfitci y x)

```
twoway lfitci outcome warmth  
  
graph export mylfitci.png, width(1500) replace
```

8 Combine Scatterplot and Linear Fit (twoway (scatter y x) (lfit y x))

```
twoway (scatter outcome warmth, mcolor(%30)) (lfit outcome warmth)  
  
graph export myscatterlinear.png, width(1500) replace
```

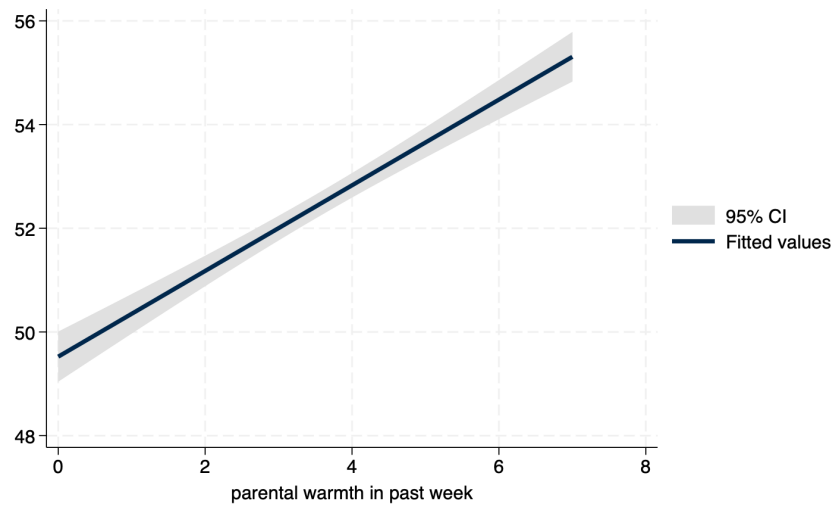


Figure 3: Linear Fit With Confidence Interval

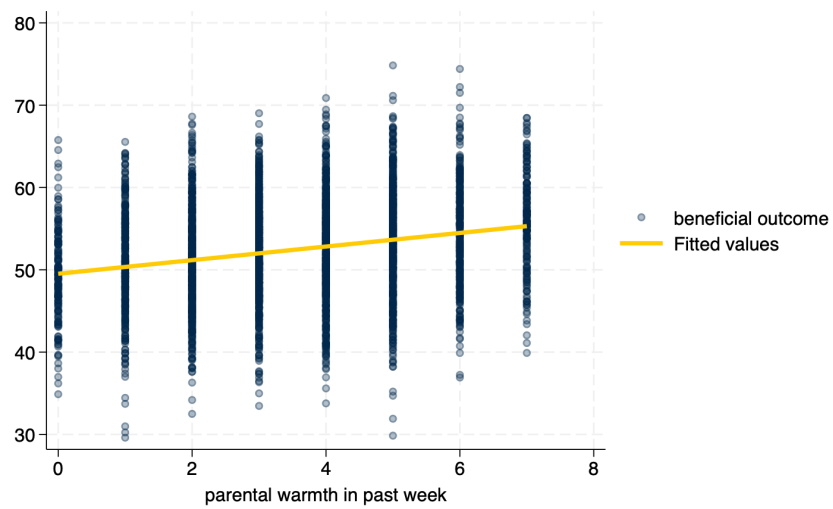


Figure 4: Scatterplot and Linear Fit

9 Spaghetti Plots (spagplot y x, id(group))

```
spagplot outcome warmth, id(country)
graph export myspaghetti.png, width(1500) replace
```

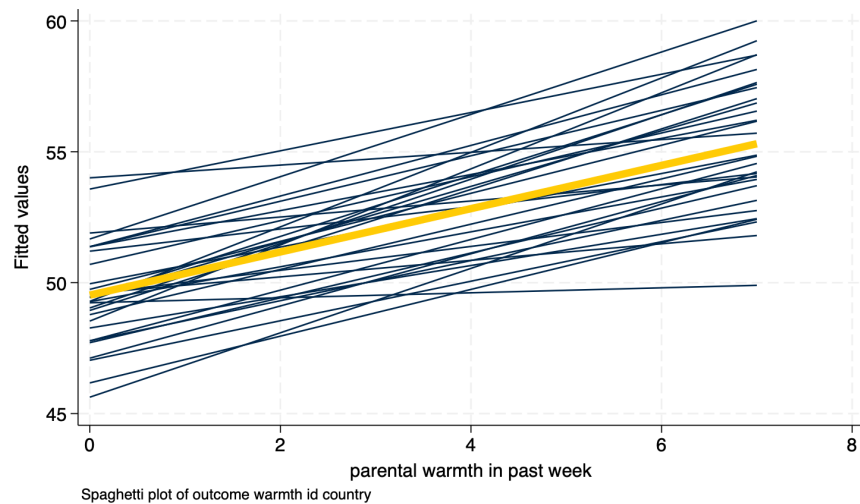


Figure 5: Spaghetti Plot

10 Small Multiples (twoway y x, by(group))

Small Multiples, showing a separate graph for each group in the data, are an increasingly popular data visualization technique. Below, I build a small multiples graph using the `by` option in Stata. I use the `aspect` option to adjust the *aspect ratio* of the graph for better visual presentation.

```
twoway (scatter outcome warmth, mcolor(%30)) ///
(lfit outcome warmth), ///
by(country) aspect(1)

graph export mysmallmultiples.png, width(1500) replace
```

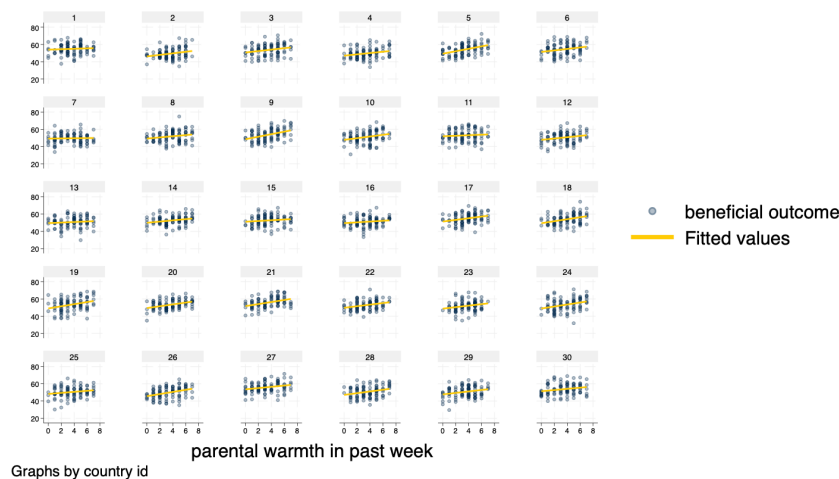


Figure 6: Small Multiples

11 Small Multiples With A Random Sample

At times, we may have *too many* Level 2 units to effectively display them on a *spaghetti plot*, or using *small multiples*. If this is the case, we may need to *randomly sample* **Level 2** units. This can be difficult to accomplish as our standard `sample` command operates on each row, or on Level 1 units.

We can accomplish random sampling at Level 2, with a little bit of code.

```
set seed 3846 // random seed for reproducibility

gen randomid = runiform() // generate a random id variable

* by country (i.e. by Level 2 unit) replace the randomid
* with the first randomid for that country (Level 2 unit)
* so that every person in that country has the same random id

bysort country: replace randomid = randomid[1]

summarize randomid // descriptive statistics for random id

twoway (scatter outcome warmth, mcolor(%30)) /// scatterplot
(lfit outcome warmth) /// linear fit
if randomid < .5, /// only use a subset of randomids
```

```
by(country) aspect(1) // by country
quietly: graph export mysmallmultiples2.png, width(1500) replace
```

(2970 real changes made)

Variable	Obs	Mean	Std. dev.	Min	Max
-----+-----					
randomid	3,000	.6174022	.2374704	.0733026	.9657055

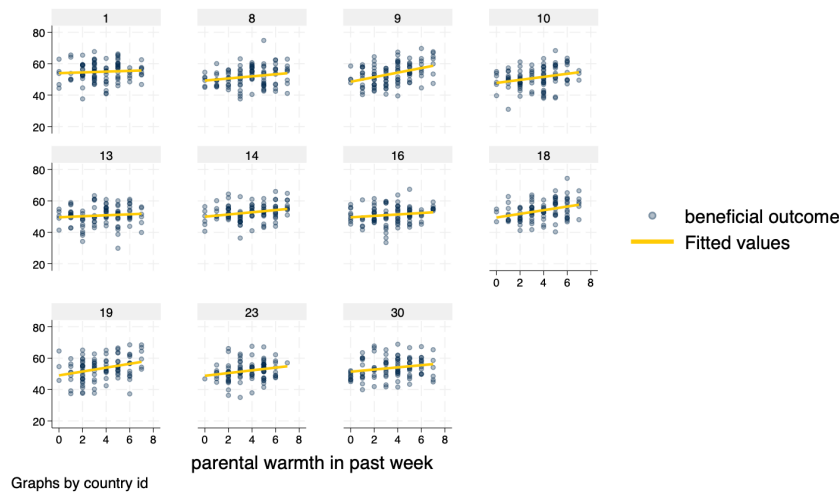


Figure 7: Small Multiples With A Random Sample Of Countries

12 Multivariate (Predicted) Relationships

💡 Multivariate vs. Bivariate Graphs

A sometimes unacknowledged point is that graphs—unless we take steps to correct this—reflect *unadjusted*, or *bivariate* associations. We may sometimes wish to develop a graphs that reflect the *adjusted* or *predicted* estimates from our models.

12.1 Using Predicted Values (predict)

`predict` generates a predicted value for *every observation in the data*.

🔥 Prediction Requires Careful Thinking

In multilevel models, *prediction* is a complex question. Prediction may—or may not—incorporate the information from the random effects. The procedures below outline graphs that incorporate predictions using the random effects, by using the `predict ...`, `fitted` syntax.

12.1.1 Estimate The Model

```
mixed outcome warmth physical_punishment i.intervention || country: // estimate MLM
```

Performing EM optimization ...

Performing gradient-based optimization:

Iteration 0: Log likelihood = -9628.1621

Iteration 1: Log likelihood = -9628.1621

Computing standard errors ...

Mixed-effects ML regression

Group variable: country

Number of obs = 3,000

Number of groups = 30

Obs per group:

min = 100

avg = 100.0

max = 100

Wald chi2(3) = 370.90

Prob > chi2 = 0.0000

Log likelihood = -9628.1621

	outcome	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
-----+-----							
	warmth	.8330937	.0574809	14.49	0.000	.7204332	.9457543
physical_punishment		-.9937819	.0798493	-12.45	0.000	-1.150284	-
.8372801							
1.intervention		.6406043	.2175496	2.94	0.003	.214215	1.066994
_cons		51.65238	.4664841	110.73	0.000	50.73809	52.56668

Random-effects parameters		Estimate	Std. err.	[95% conf. interval]	
country: Identity					
	var(_cons)	3.371762	.9613269	1.928279	5.895816
	var(Residual)	35.0675	.910002	33.32853	36.89721
LR test vs. linear model: chibar2(01) = 204.14 Prob >= chibar2 = 0.0000					

12.1.2 Generate Predicted Values

```
predict outcome_hat, fitted // predict yhat (`fitted` uses fixed AND random effects)
```

12.1.3 Graph With twoway Syntax

```
twoway (scatter outcome_hat warmth, mcolor(%30)) (lfit outcome_hat warmth)
graph export mypredictedvalues.png, width(1500) replace

twoway (lfit outcome_hat warmth)
graph export mypredictedvalues2.png, width(1500) replace
```

12.1.4 Spaghetti Plot With Predicted Values

```
spagplot outcome_hat warmth, id(country)
graph export myspaghetti2.png, width(1500) replace
```

12.2 margins and marginsplot

In contrast to `predict`, which generates a predicted value for *every observation in the data*, `margins` generates predicted values at *specific values of certain variables*.

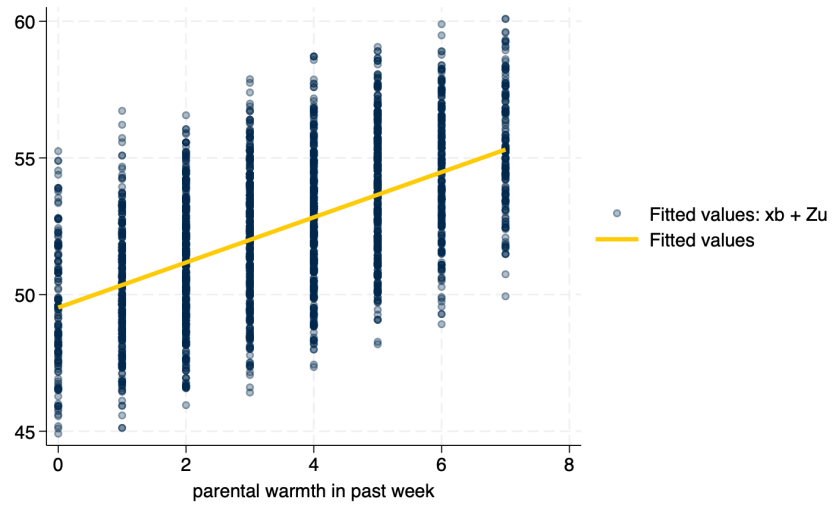


Figure 8: Predicted Values From `predict`

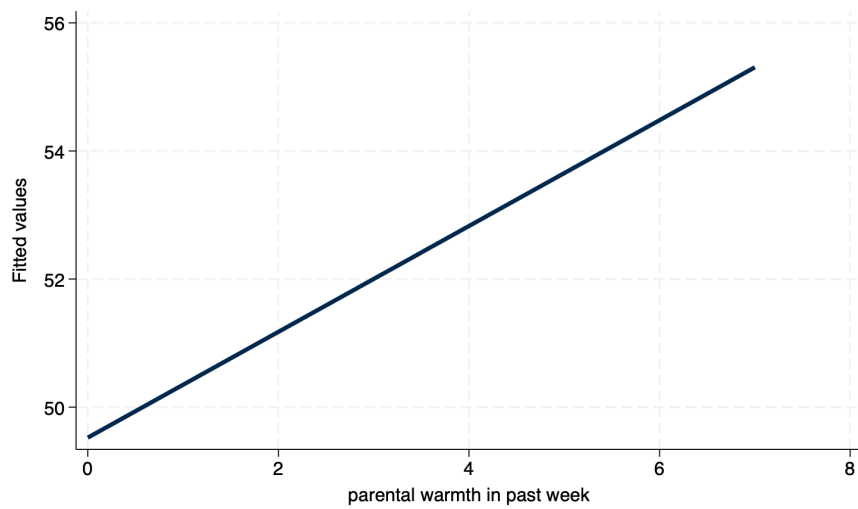


Figure 9: Predicted Values From `predict` With Only Linear Fit

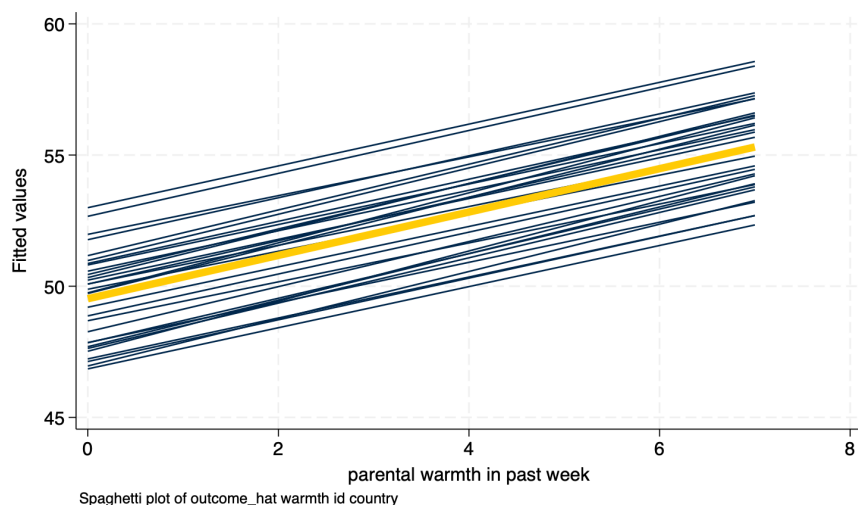


Figure 10: Spaghetti Plot With Predicted Values

12.2.1 Estimate The Model

```
mixed outcome warmth physical_punishment i.intervention || country: // estimate MLM
```

Performing EM optimization ...

Performing gradient-based optimization:

Iteration 0: Log likelihood = -9628.1621

Iteration 1: Log likelihood = -9628.1621

Computing standard errors ...

Mixed-effects ML regression

Group variable: country

Number of obs = 3,000

Number of groups = 30

Obs per group:

min = 100

avg = 100.0

max = 100

Wald chi2(3) = 370.90

Prob > chi2 = 0.0000

Log likelihood = -9628.1621

outcome	Coefficient	Std. err.	z	P> z	[95% conf. interval]
---------	-------------	-----------	---	------	----------------------

	warmth		.8330937	.0574809	14.49	0.000	.7204332	.9457543
physical_punishment		-.9937819	.0798493	-12.45	0.000	-1.150284	-	
.8372801								
1.intervention		.6406043	.2175496	2.94	0.003	.214215	1.066994	
_cons		51.65238	.4664841	110.73	0.000	50.73809	52.56668	

Random-effects parameters		Estimate	Std. err.	[95% conf. interval]	
country: Identity					
var(_cons)		3.371762	.9613269	1.928279	5.895816
var(Residual)		35.0675	.910002	33.32853	36.89721
LR test vs. linear model: chibar2(01) = 204.14 Prob >= chibar2 = 0.0000					

12.2.2 Generate Predicted Values *At Specified Values* With margins

```
margins intervention, at(warmth = (1 2 3 4 5 6 7)) // predictive *margins*
```

Predictive margins

Number of obs = 3,000

Expression: Linear prediction, fixed portion, predict()

```
1._at: warmth = 1
2._at: warmth = 2
3._at: warmth = 3
4._at: warmth = 4
5._at: warmth = 5
6._at: warmth = 6
7._at: warmth = 7
```

		Delta-method				[95% conf. interval]	
		Margin	std. err.	z	P> z		
_at#intervention							
1 0		50.02222	.3966755	126.10	0.000	49.24475	50.79969
1 1		50.66283	.3955286	128.09	0.000	49.88761	51.43805

2	0		50.85532	.3788571	134.23	0.000	50.11277	51.59786
2	1		51.49592	.3789096	135.91	0.000	50.75327	52.23857
3	0		51.68841	.3692182	139.99	0.000	50.96476	52.41207
3	1		52.32902	.370554	141.22	0.000	51.60274	53.05529
4	0		52.52151	.3684014	142.57	0.000	51.79945	53.24356
4	1		53.16211	.3710204	143.29	0.000	52.43492	53.8893
5	0		53.3546	.376464	141.73	0.000	52.61674	54.09246
5	1		53.9952	.3802764	141.99	0.000	53.24988	54.74053
6	0		54.18769	.3928599	137.93	0.000	53.4177	54.95768
6	1		54.8283	.3977088	137.86	0.000	54.0488	55.60779
7	0		55.02079	.4166062	132.07	0.000	54.20425	55.83732
7	1		55.66139	.4223062	131.80	0.000	54.83369	56.4891

12.2.3 Graph With marginsplot

```
marginsplot // plot of predicted values
graph export mymarginsplot.png, width(1500) replace
```

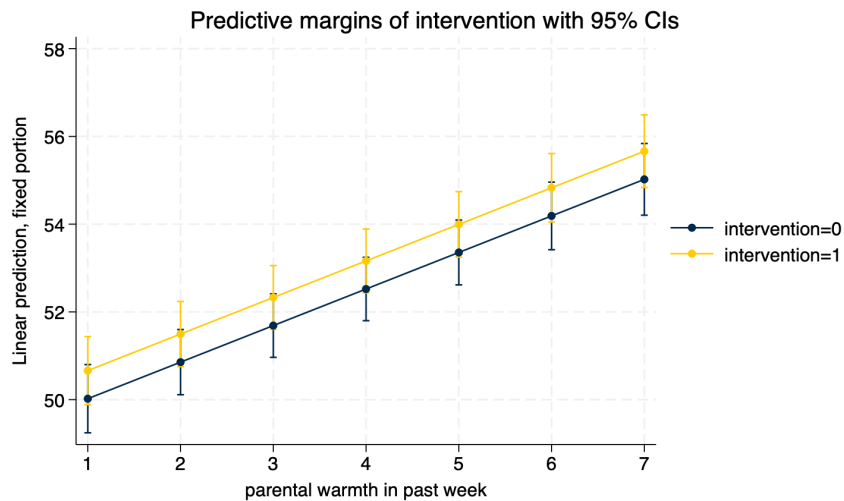


Figure 11: Predicted Values From `margins` and `marginsplot`

13 Scatterplot With Linear Fit and Marginal Density Plots (twoway ...)

As another possibility, we may wish to show more of the variation, by showing the variation in the *independent* variable and the *dependent* variable along with a *scatterplot* and *linear fit*. This is a complex graph and requires a little bit of manual programming in Stata.

i binscatterhist

You could also investigate the user written program `binscatterhist` (`ssc install binscatterhist`) which produces a similar looking graph, and automates much of this work.

13.1 Manually Generate The Densities To Plot Them Below (`kdensity ...`)

i Note

We generate the density for *warmth* at only a few points (`n(8)`) since this variable has relatively few categories.

```
kdensity warmth, generate(warmth_x warmth_d) n(8) // manually generate outcome densities
kdensity outcome, generate(outcome_y outcome_d) // manually generate outcome densities
```

13.2 Rescale The Densities So They Plot Well

i Note

You may have to experiment with the scaling and moving factors.

```
replace warmth_d = 100 * warmth_d // rescale the density so it plots well
replace outcome_d = 5 * outcome_d - .5 // rescale AND MOVE the density so it plots well
label variable outcome_y "density: beneficial outcome" // relabel y variable
```

(8 real changes made)

(50 real changes made)

13.3 Make The Graph (twoway ...)

i Note

You may have to experiment with whether scatterplots or line plots work best for displaying the x and y densities.

```
twoway (scatter outcome warmth, mcolor(%10)) /// scatterplot w some transparency
(lfit outcome warmth) /// linear fit
(line warmth_d warmth_x) /// line plot of x density
(line outcome_y outcome_d), /// line plot of y density (note flipped order)
title("Outcome by Warmth") /// title
ytitle("beneficial outcome") /// manual ytitle
xtitle("parental warmth") /// manual xtitle
legend(position(6) rows(2) ) /// legend at bottom; 2 rows
xlabel(0 1 2 3 4 5 6 7) /// manual x labels
name(mynewscatter, replace)

graph export mynewscatter.png, width(1500) replace
```

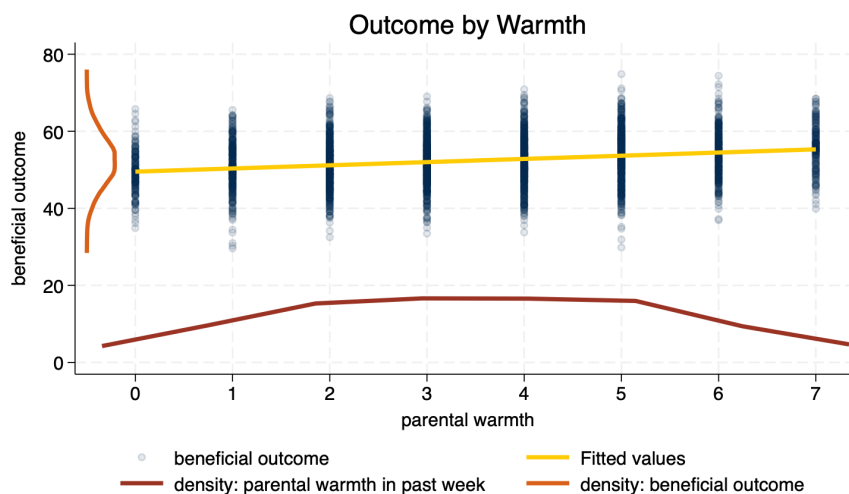


Figure 12: Scatterplot and Linear Fit With Marginal Density Plots

References

Hemken, Doug. 2023. *Statamarkdown: 'Stata' Markdown*. <https://CRAN.R-project.org/package=Statamarkdown>.